



Computer Engineering Department

Course Name: Microprocessor Lab

Number: 10636392

Lab Report Grading Sheet

Instructor : Dr. Aladdin Masri	Experiment # : 7
Academic Year : 2025 - 2026	Experiment Name: Memory Read/Write
Semester : second semester	

Students 1 – Aseel Qadan			2 – Aryam Sous	
Performed on : 10/5/2026			Submitted on : 15/5/2026	
Report's Outcomes				
ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %
Evaluation Criterion			Grade	Points
Abstract answers of the questions: “What did you do? How did you do it? What did you find?” Introduction and Theory Sufficient, clear and complete statement of objectives. In addition to Presents sufficiently the theoretical basis. Apparatus/ Procedure Apparatus sufficiently described to enable another experimenter to identify the equipment needed to conduct the experiment.Procedure sufficiently described.			0.5	
Experimental Results and Discussion (In-Lab Worksheet) Crisp explanation of experimental results. Comparison of theoretical predictions to experimental results, including discussion of accuracy and error analysis in some cases.			2	
Conclusions and Recommendations Conclusions summarize the major findings from the experimental results with adequate specificity. Recommendations appropriate in light of conclusions. Correct grammar.			4	
Appearance Title page is complete, page numbers applied, content is well organized, correct spelling, fonts are consistent, good visual appeal.			1	
			1	
Total			10	



Objectives

This experiment aimed to explore the practical applications of static RAM and to master the programming methods for reading and writing data associated with it. In digital systems, storage blocks play a crucial role, and this lab focused on retrieving data stored in static RAM. The memory contained values that were subsequently used to control the speed of a DC motor.

Introduction

Digital systems commonly require storage, and in this lab, we investigated the use of static RAM (SRAM). The memory module employed in the experiment held data that was later utilized to control the speed of a DC motor.

Procedure:

o In Part 1, we stored four values (00, 06h, 0Bh, 0Fh) in the SRAM memory, representing the speed of the DC motor. These values were assigned to addresses 100, 101, 102, and 103, respectively.

o Part 1 Code:

```
CODE SEGMENT
ASSUME CS:CODE
ORG 2000H

START:
mov bx,100h
mov ax,0
mov ds,ax

mov cx,4
mov si,offset array

lp:
mov al,[si]
mov ds:[bx],al
inc bx
inc si
loop lp

jmp START

array db 00h,60h,0B0h,0F0h

END START
CODE ENDS
```

0100	00	60	0B	0F	15	A2	55	8A	51	A2	15	AA	64	A3	55	32
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The second phase introduces button-driven speed regulation. Pressing any button instructs the program to reference a unique memory location—such as 100H for the first input, 101H for the second, and so on. The data fetched from that specific address acts as a configuration value to dictate the motor's operating velocity.

The code part 2

```
CODE SEGMENT
ASSUME CS:CODE
ORG 1000H

START:
mov dx,0ff2bh
mov al,90h
out dx,al

mov dx,0ff80h
mov al,0
out dx,al

mov bx,100h
mov ax,0
mov ds,ax

mov si,offset array
mov cx,4

l1:
mov al,[si]
mov ds:[bx],al
inc si
inc bx
loop l1
```

```
l2:
mov dx,0ff28h
in al,dx

mov cx,0ffh
lpp:
nop
loop lpp

or al,11100000b
cmp al,11111110b
je button1
cmp al,11111101b
je button2
cmp al,11111011b
je button3
cmp al,11110111b
je button4
jmp l2

button1:
mov bx,0100h
mov al,ds:[bx]
mov dx,0ff80h
out dx,al
jmp l2

button2:
```

```
button2:
mov bx,0101h
mov al,ds:[bx]
mov dx,0ff80h
out dx,al
jmp l2

button3:
mov bx,0102h
mov al,ds:[bx]
mov dx,0ff80h
out dx,al
jmp l2

button4:
mov bx,0103h
mov al,ds:[bx]
mov dx,0ff80h
out dx,al
jmp l2

array db 00h, 60h, 0B0h, 0F0h
END START
CODE ENDS
```



Conclusion

This experiment provided us with hands-on experience in working with memory, deepened our understanding of the applications of static RAM, and demonstrated how to control the speed of a DC motor using Memory Read operations.